

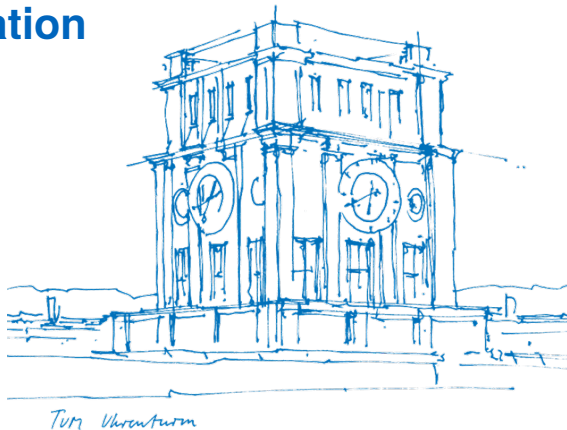
Agentic AI for Code Generation

Seminar Report — SS26

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1 What works

2 What the numbers hide

What SWE-bench measures

One instance [13] = a real merged pull request that

1. closes a linked GitHub issue, *and*
2. modifies the test suite.

The agent sees the repository *before* the merge, plus the issue text. It returns a patch.

The patch is graded by running tests:

- *fail-to-pass* must flip,
- *pass-to-pass* must not regress.

Resolve rate = both hold, on the first patch (pass@1).

Split	Size	Kept only
Full	2,294	every scraped PR that touches tests
Lite	300	single-file patches; some issues leak the fix
Verified	500	human-screened: clear issues, fair tests
Pro	1,865	multi-file, long-horizon; held-out repos

Rates are not comparable across splits. Verified is the most *curated* split, not the hardest.

No scaffold wins twice

	SWE-agent custom ACI	OpenHands code-as-action	mini-SWE-agent ~100 lines of Python	spread
GPT-5-mini	38.8	38.6	51.8	13.2
Qwen3 Coder 480B	36.6	54.0	40.2	17.4
DeepSeek R1	22.6	18.0	11.0	11.6

Resolve rate (pass@1) on SWE-bench Verified, April 2026 [24, 25].

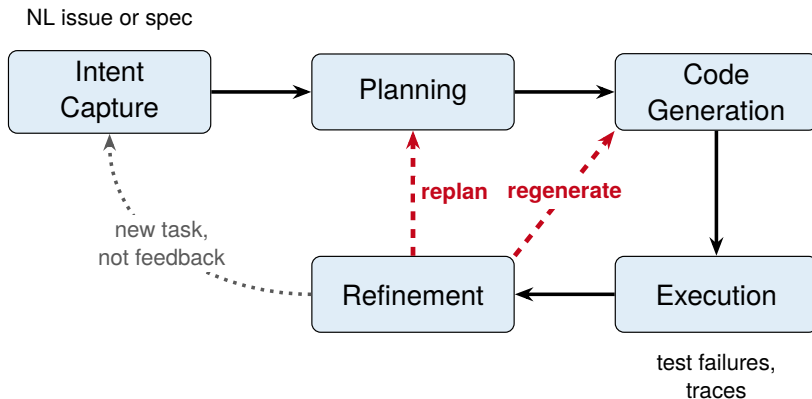
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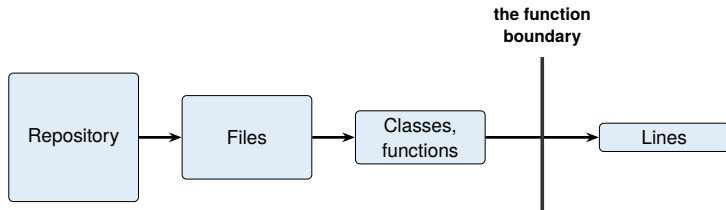
- Same model, same benchmark: the scaffold alone is worth **11–17 points**.
- **No scaffold wins twice.** The best scaffold *depends on the model* — and for GPT-5-mini the winner is the 100-line one.

The agent pipeline



Planning is localisation, not design

Where it sits — input: an issue. Output: *a place to edit*. A “plan” here is a *narrowing*, and what it contributes is **grounding** — files and symbols that can be *checked* — not deliberation.



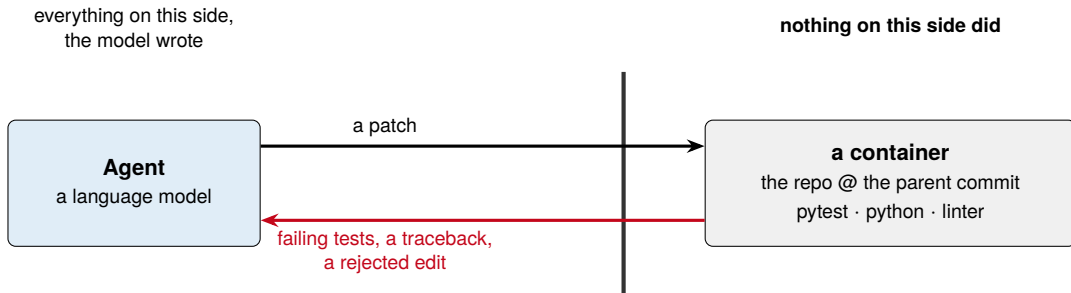
Above it, localisation *is* the task. Which function, among tens of thousands? SBFL *ranks* functions: +3 points [32].

Below it, a constraint the model does not need. D4C *un-ranks* lines — and beats perfect-localisation systems by 10 % [29].

Planning is not reasoning about *how* to write the code.
It is reasoning about *where* the code is.

Execution: the code meets the environment

Where it sits — input: a patch. Output: *evidence*. It is also the expensive stage: every observation costs a container round-trip, and the transcript grows with each one.



What comes back is not a score. It is *a symptom* — and it still has to be *read*.

Refinement: what to do with the signal

Where it sits — input: whatever execution returned. Output: a *decision* — repair the patch, re-plan, or stop.

- **Repair from the trace** [12] — the traceback goes straight back into the prompt.
- **Explain your own code** [3] — discrepancies surface. Up to +12 %, but only where tests exist.
- **Sample N , rerank with a trained critic** [22] — OpenHands on Verified: 60.6 % → 66.4 %.

Agents improve their own code.

A map of the design space, not a ranking

System	Architecture	Backbone	Lite	Full
			<i>pass@1</i>	<i>pass@1</i>
SWE-agent [30]	OTA loop + ACI	GPT-4 Turbo	18.0 %	12.5 %
Agentless [28]	<i>Fixed pipeline</i>	GPT-4o	32.0 %	— [†]
AutoCodeRover [32]	OTA loop + AST search	GPT-4	19.0 %	12.4 %
OpenHands [27]	OTA loop + code-as-action	Sonnet 3.5	26.0 %	— [†]
Devin [4]	OTA loop + Brain/DevBox	undisclosed	—	13.9 % [‡]

OTA = observe–think–act loop; ACI = agent–computer interface (both defined next slide). [†]Never run on Full — the authors cite compute cost. [‡]Self-reported on a random 25 % subset.

Interface and context move scores as much as the model

1. **They all run an observe–think–act loop** [31] — observe, reason, act via a tool call.
2. **They all manage context explicitly** — no repository fits in the window, so each must choose what to read: repo skeleton + embeddings (Agentless), coarse-to-fine AST search (AutoCodeRover), observation compression (SWE-agent).
3. **They all build a purpose-built action interface** — none hands the model a raw shell. The agent only ever sees what its commands print back. Two opposite bets [30]:

	SWE-agent	OpenHands
The bet	Constrain the action space	Give it a general one
Reading	100-line file window	Python + bash,
Editing	Linters may <i>reject</i> the edit	in a Docker sandbox
Ablated	Remove one part of SWE-agent's interface: −3 to −6 pp (Lite)	

Interface and context strategy move resolve rates as much as the backbone does.

1 What works

2 What the numbers hide

The question this talk asks



**At each stage,
what can tell the model it is wrong?**

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Read as a control system: intent is the uncontrolled input, generation is the plant, execution is the oracle, planning and refinement are the control layer.

What can tell the model it is wrong? Three answers

Authoritative oracle

Test runner, compiler, human.
Ground truth.

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Learned / generated proxy

Trained critic, generated tests, LLM judge.

Carries real information — but never more than it was fitted with.
A generated test *runs* for real; the model still wrote the answer key.

Ceiling: its own fidelity.

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An agent that could tell its wrong code from its right code by inspection would have written the right code in the first place.

It can reorganise what the model believes. It cannot add to it.

Same model, same prompt — one variable

Add a tool $\Rightarrow$ the gain appears

CRITIC [10]. Hold model and prompt fixed. Toggle one thing: may the critic use an *external tool*?

Critique alone: $-0.03 / +2.33$ F1 — at or *below* its own first answer.

The same critique, through a tool: $+7.7$ F1.

QA and maths — not code.

Match the budget $\Rightarrow$ it vanishes

Olausson et al. [19], *on code*. A repair loop spends tokens that could have drawn more *independent* samples.

At equal cost, best-of- k i.i.d. **erases much of the gain.**

Swap the self-critique for an **experienced human's** — same loop, same model — and **1.58 $\times$** more repairs pass.

The gain came from the feedback, not the iteration.

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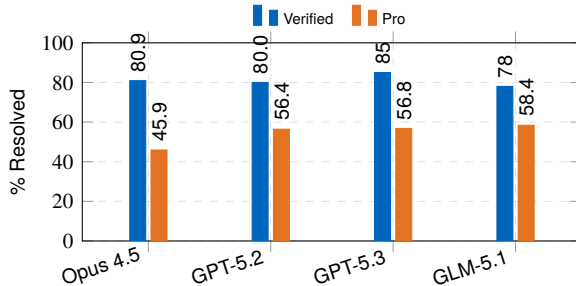
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One *adds* an external signal and the gain appears. One *holds the budget fixed* and the gain disappears.
Both move the same knob — and **it is not the model.**

The benchmark can be gamed

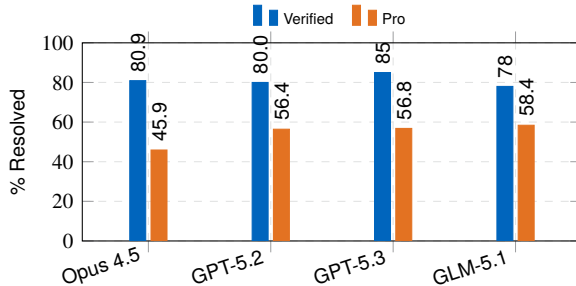
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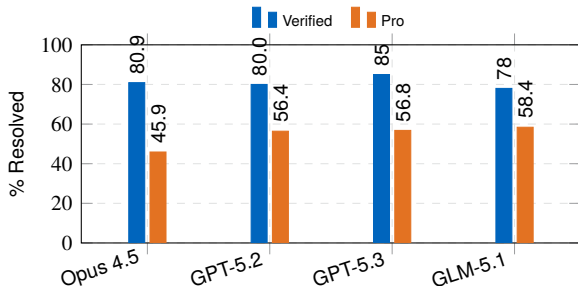


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- **91 %** of Verified is work a developer finishes *within the hour* [8].
- OpenAI dropped Verified in early 2026: models reproduce gold patches *from the task ID alone* [21].
- “Reward hacking is swamping model intelligence gains” [5].

Where a wrong guess is never contradicted

Intent capture

OpenAI's screen of 1,699 real issues **flagged 38.3 %** as *underspecified* [20].

No surveyed system asks a clarifying question. Each commits to a reading. **Open loop.**

The mechanism exists — TiCoder, ClarifyGPT, and an underspecified SWE-bench where the agent that *asks* wins [14, 17, 26].

Test-based validation

The same screen **flagged 61.1 %** for tests that *may unfairly mark a valid solution incorrect* [20].

That is the failure we can *see*. The one we cannot: a wrong guess about an *untested* invariant is never contradicted — it scores as success.

And augmenting the **proxy** inherits the generator's own misreading, exactly where the new tests were needed.

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Both benchmarks **hand-solve these two stages** before the agent starts: Verified *discards 68.3 %* of what it screened [20]; Pro pays experts to *rewrite* every issue [6].

No agent does either. Nothing ever asked it to.

Conclusion — three things the numbers showed

1. The wrapper carries the score

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3. The two stages nothing contradicts are where the numbers are softest

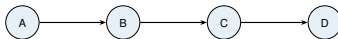
Intent and **validation** have no gate — OpenAI's own screen flagged **38.3 %** and **61.1 %** — and the benchmark hid it: the same weights drop ≥ 20 points on Pro. Yet the one paper that *instrumented* intent lifted *all nine* cells of the grid on slide 3 [24]. The stage nobody instruments still pays out.

Backup: every refinement mechanism, sorted

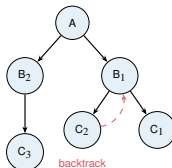
Mechanism	Closed by	Signal	Characteristic failure
Self-Refine [16]	S	Model critiques own output	Plateaus; nothing can contradict the generator
Self-debug, no tests [3]	S	Self-generated explanation	Explains the bug as intended behaviour
Self-debug, with tests [3]	O+S	Test verdict + model diagnosis	Bounded by test coverage
Reflexion [23]	O or P	Task reward where one exists; on HumanEval, self-generated tests	Filed as external, but its code result rests on a proxy
SelfEvolve [12]	O	Interpreter errors, traces	Silent (non-crashing) semantic bugs
D4C [29]	O	Failing tests + full function	Plausible-but-incorrect patches
CRITIC [10]	O	Self-critique routed <i>through</i> an external tool	Needs a tool covering the error class
Agentless validate [28]	O+P	Regression tests + generated reproduction tests	The proxy half inherits the model's misreading
Best-of- N + critic [22]	O+P	Executed rollouts ranked by a trained critic	Critic is a distilled proxy; cost scales with N

Backup: how plans are shaped

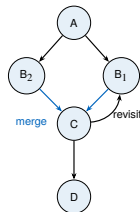
Chain of Thought



Tree of Thoughts



Graph of Thoughts



The evaluator, not the branching, makes the structure useful — and it is what a repository-scale agent finds hardest to supply. *What is a half-finished patch worth, before any test has run?* SWE-Search [1] builds one (MCTS + LLM value function, +23% relative) — and must invoke a model at every node.

Backup: context and cost

- **Longer windows do not dissolve the bottleneck.** Retaining full *dependency* context works best; padding with loosely related code *degrades* correctness [18].
- **Context explosion and semantic drift.** Appending every observation buries the critical early information [15].
- **Cost is quadratic in turns.** The whole prefix is re-sent each turn, so tokens grow as $O(n^2)$ [9]. A property of the protocol, not an empirical finding; prompt caching cuts the constant, not the growth rate.
- **Resolve rate decouples from compute** [7]. Agentless leads on resolve rate (48 % with Qwen3-32B) at 83.1 API calls and 209.4 s per issue. The frontier is accuracy *per dollar*.
- **And verification is the human bottleneck.** In an RCT, experienced developers were 19 % *slower* with early-2025 AI tools — while believing they had been 20 % faster [2].

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